

Switching algorithms of a CC-CV strategy for battery charging of electric vehicles

Salma Fadili^{1,2}, Alain Bouscayrol¹, Eric Noirat², Philippe Delarue¹, Philippe Fiani², Clément Mayet¹

¹Univ. Lille, Arts et Métiers Institute of Technology, Centrale Lille, Junia, ULR 2697 - L2EP, F-59000 Lille, France

²Sherpa engineering, F-92000 Nanterre, France

Corresponding author: Alain.Bouscayrol@univ-lille.fr

Abstract — This study examines the Constant Current-Constant Voltage (CC-CV) charging strategy for batteries of electric vehicles. The charging subsystem model and control are organised using the Energetic Macroscopic Representation formalism. The coupling of a CC control loop and a CV control loop is insured by a switching algorithm. Various algorithms are explored: a binary switching, a progressive switching and a current limitation technique. Despite its simplicity, the binary switching leads to some current peaks in the battery. The two other switching algorithms avoid this drawback. The charging time remains the same for all the switching algorithms.

Keywords — Battery charging, CC-CV, electric vehicles, energetic macroscopic representation

I. INTRODUCTION

Batteries of Electric Vehicles rely on a classical charging strategy known as Constant Current-Constant Voltage (CC-CV) [1] [2]. Other strategies are developed such as CP-CV (Constant Power Constant Voltage) or pulse charge, etc. But they are not yet widely used [3]. The CC-CV strategy allows efficient charging, ensures battery safety, optimises battery lifespan and manages temperature. Two operation modes must be considered. First, in the CC (Constant Current) mode, a maximal current is imposed to charge the battery during a short time. When the final battery voltage is reached, a voltage regulation is imposed to progressively lower the current avoiding a voltage drop that leads to a Constant Voltage (CV) [4]. The CC-CV strategy is also used in optimization of the EV charging infrastructure [5].

In the CC-CV strategy, the control must thus switch from a CC to a CV mode. The switching implementation is not always described and different solutions can be developed. A first control organization has been proposed based on EMR (Energetic Macroscopic Representation) using a switching element between a CC control loop and a CV control loop [6]. EMR is a graphical formalism to organize model and control of energy conversion systems [7]. In that methodology, the control organization is derived from the model organization. The control scheme deduced from EMR proposes a clear organization of this switching operation. However, some current peaks appear during the switching.

The objective of this paper is to study other switching algorithms in order to avoid current peaks. EMR is used to organize different control schemes. Different solutions are compared.

Section II presents the studied system and its EMR. Section III presents the control structures deduced from the EMR. Section IV presents the different strategies used and the results obtained for each method.

II. EMR OF THE CHARGING SYSTEM

A. Studied charging system

In this study, the focus is on the analysis of the charging subsystem of an external (off-board) charger, an essential component within the framework of electric vehicles infrastructure. The charger is composed of three distinct subsystems: the supply subsystem powered from the grid, the charging subsystem responsible for managing the DC bus and the connection to the EV [8].

However, in this study, the focus will only be on the charging subsystem (Fig. 1). This part includes the DC bus, the chopper, the inductor filter, and the connection with the battery. The subsystem parameters correspond to the parameters of the EVbox Troniq 50, while the battery parameters correspond to the 2018 Nissan Leaf battery (Table 1).

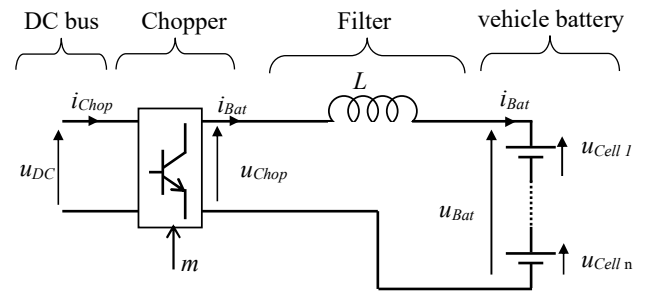


Fig. 1. Structural scheme of the charging subsystem

Table 1. Technical parameters

Battery	ASC-Envision NMC
Energy	40 kWh
Number of modules	24
Nominal voltage	350 V
Capacity	112 Ah
Charging station	EVbox Troniq 50
Charging power	50 kW
DC Bus	650 V
Maximal charging current	120 A

B. EMR of the studied system

In this paper, all elements of the studied system are represented by the EMR formalism.

EMR serves as a highly compatible tool to organise models of energy conversion systems [7]. Rooted in the fundamental principles of interaction and causality, EMR offers a structured approach for modelling and control energy conversion systems. In the context of this study, EMR is used to organise and control the charging subsystem (Fig. 2).

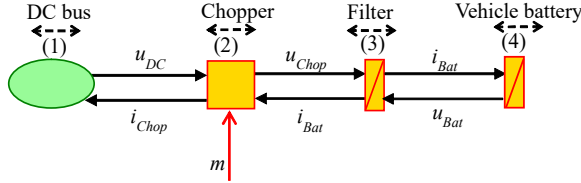


Fig. 2. EMR of the charging subsystem connected to the battery

Using the action-reaction principle, EMR describes each subsystem with a specific shape and colour depending on its function (see Appendix). The studied system is depicted by orange pictograms. The external elements are depicted by green pictograms. The control is depicted by blue pictograms. The controlled DC bus is considered as a constant voltage source. Energy sources are represented with green oval pictograms. The chopper is a DC-DC converter that adjusts the voltage from the DC bus to the voltage needed to charge the battery. In EMR, a chopper is a mono-domain conversion element represented with an orange square. The filter is used with a chopper to reduce the ripple effect on the current. The filter is considered an accumulation element due to its internal energy storage. In EMR, it is represented with a crossed orange rectangle. The battery stores the energy received from the charging circuit. The battery is also an accumulation element (crossed orange rectangle).

The DC bus is a constant voltage source u_{DC} :

$$u_{DC} = 650 \text{ V} = \text{cst} \quad (1)$$

The chopper is modelled in average value with m the modulation ratio, η the chopper efficiency, u_{chop} and i_{chop} , the voltage and current chopper and i_{Bat} the filter current that is also the battery current:

$$\begin{cases} u_{chop} = m u_{DC} \\ i_{chop} = \eta m i_{Bat} \end{cases} \quad (2)$$

The filter is modelled by the classical relationship with L its inductance, R its resistance, u_{bat} the battery voltage:

$$i_{Bat} = \frac{1}{L} \int_0^t (u_{chop} - u_{Bat} - R i_{Bat}) dt \quad (3)$$

The battery is only modelled by its State of Charge (SoC), with C_{Ah} its capacity:

$$SoC_{Bat} = SoC_{init} + \frac{1}{3600 C_{Ah}} \int_0^t i_{Bat} dt \quad (4)$$

III. INVERSION-BASED CONTROL STRUCTURE

In the CC-CV strategy, a first mode (CC) enables a constant charge with a constant current. A second mode (CV) reduces progressively the current by imposing a constant voltage (Fig. 3). A switching algorithm (Fig. 4) must be defined to switch between the CC and the CV modes when the battery is close to the final SoC [9].

A way to define the global control scheme is to define the control of each model first, and couple then in a second time. EMR is used in that aim.

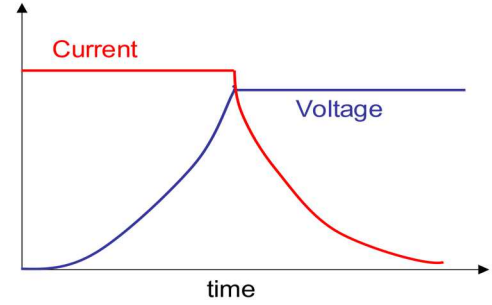


Fig. 3. CC-CV strategy

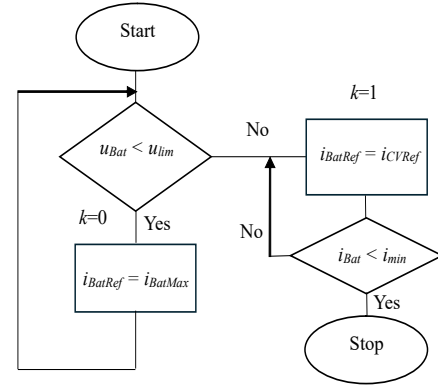


Fig. 4. Flowchart of the CC-CV strategy

A. Control of the CC mode

In the CC mode, the charging process begins by imposing a constant current to the battery. The current level is determined by the charging system, ensuring a rapid initial charge. During this stage, the battery state of charge increases progressively as it absorbs energy.

The related control scheme can be defined by an inversion of the tuning path of the EMR, from the modulation ratio m to the battery current i_{bat} (Fig. 5). The control scheme is thus derived from inversion rules (Fig. 6).

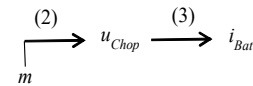


Fig. 5. Tuning path of the CC mode

The reference current i_{CCRef} is imposed by a strategy and can be the maximal battery current to enable the faster charging:

$$i_{CCRef} = i_{BatMax} \quad (5)$$

The first equation to invert is (3). As it is related to an accumulation element with delay between input and output, the EMR rules indicate that this inversion is ensured by a closed-loop control:

$$u_{ChopRef} = C_{pi}(t)(i_{CCRef} - i_{BatMeas}) + u_{BatMeas} \quad (6)$$

with $C_{pi}(t)$ a controller that can be PI or other. This control loop delivers the reference of the chopper voltage $u_{ChopRef}$. The second relationship to invert is (2). As there is no delay between input and output, a direct inversion is possible:

$$m = \frac{u_{ChopRef}}{u_{DCMeas}} \quad (7)$$

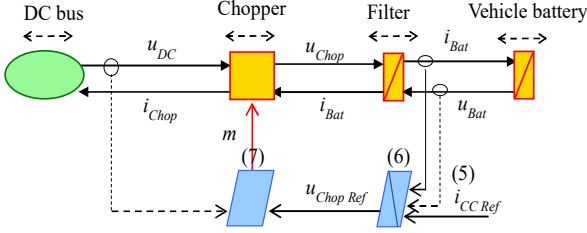


Fig. 6. EMR and control in the CC mode

B. Control of the CV mode

In the CV mode, the charging voltage remains constant, while the charging current gradually decreases. This phase occurs when the battery approaches its maximal charge capacity. During this stage, the decreasing current prevents overcharging by limiting the voltage applied to the battery terminals.

The control scheme is also derived from an inversion of the tuning path of the EMR from the modulation function m to the battery voltage u_{bat} (Fig. 7). The control scheme is thus derived from these inversions (Fig. 8).

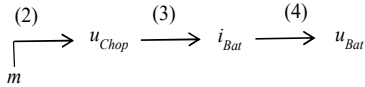


Fig. 7. Tuning path of the CV mode

The final voltage is considered as the reference:

$$u_{BatRef} = u_{BatMax} \quad (8)$$

The first equation to invert is (4). As the battery is described by an accumulation element, another control loop is needed for this inversion:

$$i_{CVRef} = C_p(t)(u_{BatRef} - u_{Bat}) \quad (9)$$

With $C_p(t)$ a controller that can be P or other. This control loop delivers the reference of the filter current i_{CVRef} . In the case of a P controller, this current reference is directly proportional to the difference between the voltage and its reference. As a positive battery current lead to increase the battery voltage, the voltage difference decreases and thus the reference current. This process leads to a slow decrease of the

battery current to avoid a current step and thus a voltage drop. This new reference is used as input of the inversion of (3) in the CV mode instead of (5) in the CC mode:

$$i_{CCRef} = i_{CVRef} \quad (10)$$

Of course, the inversion of (3) and (2) are realized in the same way than for the CV mode.

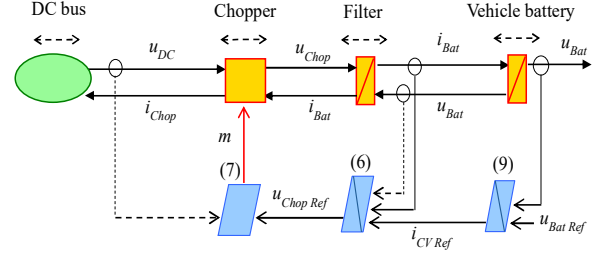


Fig. 8. EMR and control in the CV mode

C. Merging both controls

It can be observed that both controls have a common part: the inversion of (2) and the inversion of (3). A merged control can be developed by switching from the filter current reference of the CV mode (i_{BatMax}) to the one provided by the CV mode (i_{CVRef}). A coupling element is thus considered to couple both control schemes [10] (Fig. 9).

A switching algorithm is now required from a strategy to switch from the CC to the CV mode at the right time.

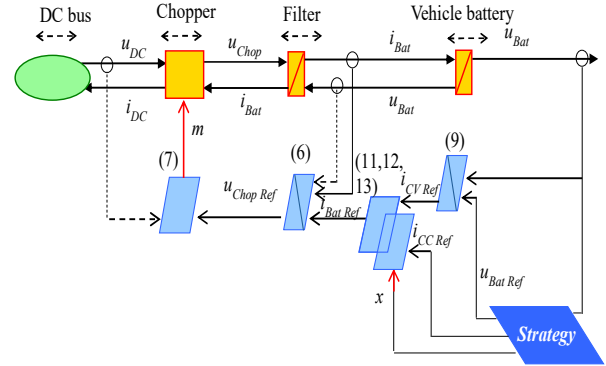


Fig. 9. Merged control including both modes

IV. DIFFERENT SWITCHING ALGORITHMS

To switch between the modes, a distribution variable is determined. Three different algorithms are explored. First, a binary switching is used between the two modes as already proposed in [6]. Second, a progressive switching is used. Finally, another algorithm is proposed with the limitation of the battery current.

A. Different algorithms

In the first algorithm, a switching equation is introduced:

$$i_{BatRef} = i_{CVRef}k + i_{CCRef}(1 - k) \quad (11)$$

With k a distribution input with 2 values: $k \in \{0, 1\}$. When the switching condition is reached, the distribution input goes

from 0 (CC mode) to 1 (CV mode). Thus, there is a step in the change of the current reference.

To avoid this current step, a continuous value of the distribution input can be used $k \in [0,1]$ with the same equation. A first order filter is added on the distribution input to enable a progressive change of its value. This operation leads to smooth and progressive change of the current reference.

In the last case, there is no switching algorithm, but a current limitation after the controller (9). At the beginning, as the voltage is far from its reference, the voltage controller leads to a very high current that is greater than the maximal battery current. A limitation is then ensured to impose the maximal battery current. This limitation indirectly leads to a CC mode with a constant current. When the voltage is close to the final value, the controller reduces the current reference. When the current is lower than the maximal battery current, the limitation is stopped and the CV mode is ensured. The limitation algorithm is ensured by the following equations:

$$i_{CV} \geq i_{BatMax}, i_{BatRef} = i_{BatMax} \quad (12)$$

$$i_{CV} < i_{BatMax}, i_{BatRef} = i_{CV} \quad (13)$$

It can be noted that as a P controller is used, no anti-windup algorithm is needed. But, if a PI controller is used, an anti-windup algorithm must be added.

B. Simulation results

The considered test is related to the charge of the Nissan leaf battery from a SoC of 20% to a SoC of 100%. This interval corresponds to the recommended interval of battery charging. Preventing the battery from falling below 20% of SoC prolongs its lifespan as deep discharges cause a bigger capacity reduction over time [11].

The 3 different algorithms are simulated and compared. They globally lead to the same evolutions of the current and voltage as indicated for the second strategy (Fig. 10).

The first algorithm with a binary k shows a current drop when the switching (Fig. 11). This is due to the switch being binary. A full charge from 20% to 100% takes 55 mins using the maximal battery current.

The second algorithm fixes the current drop noticed at first (Fig. 11). This is because the switch flips smoothly and progressively rather than instantly, thus giving enough time for the interchange to happen. A full charge of 20% to 100% takes 55 mins.

The third algorithm limits the CV current to the maximal current the battery can handle (Fig. 13). Thus, lowering the current when the charging capacity is nearing the maximum. A full charge of 20% to 100% takes 55 mins.

In terms of charging time, it can be stated that all algorithms lead to the same results. However, the binary strategy leads to current peaks that can be damageable for the battery.

In terms of complexity, the binary switching is the simpler because only a binary test is needed. The limitation algorithm is

also simple, but when a PI controller is used an anti-wind-up algorithm must be added.

In terms of flexibility, the binary and the progressive algorithms enable to consider various current profiles. They can be used for other charging strategies such as CP-CV or PC modes. For the limitation algorithm, a variable limitation must be introduced in order to consider the new reference profile.

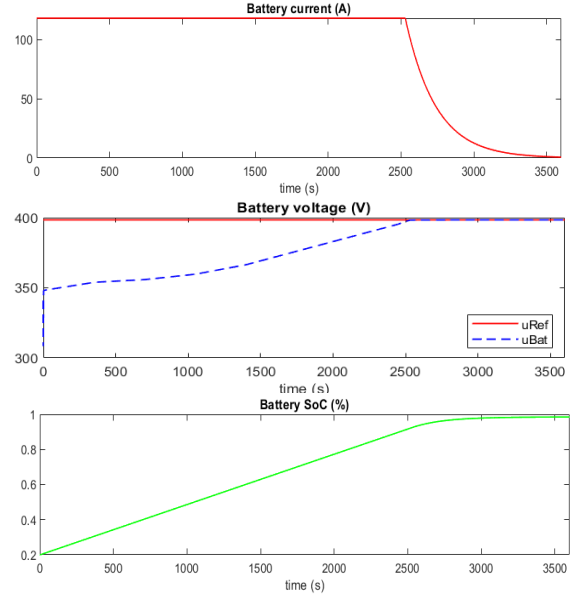


Fig. 10. Simulation results of the second algorithm

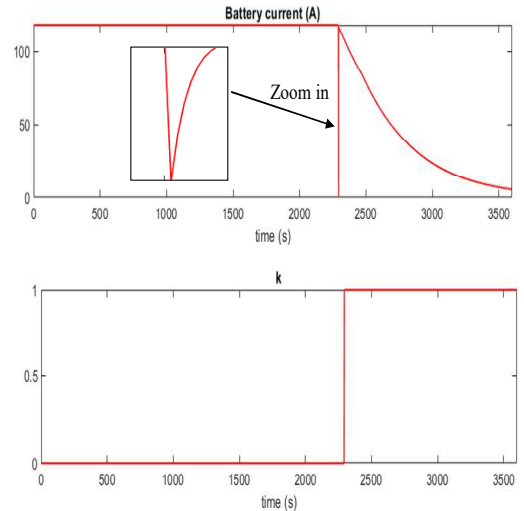


Fig. 11. First algorithm: battery current and distribution input

V. CONCLUSION

The CC-CV strategy is the most used method to charge batteries of electrical vehicles. It is composed of two modes (CC and CV) and at the end of the charging process, the control must switch from the CC mode to the CV mode. The EMR formalism has been used to describe the control of charging infrastructures to enable a CC-CV strategy.

Three switching algorithms have been compared for a dedicated charging station (EVbox Troniq 50) and a specific

car (Nissan Leaf, NMC li-ion battery). The three switching algorithms lead to the same charging time. However, if the binary switching is the simplest one, it leads to some current peaks that can impact the battery ageing. The progressive switching or the limitation algorithm should be preferred to avoid this current peaks.

Other comparisons must be made to analyse the efficiency, flexibility, the robustness and the easy implementation of these strategies. An experimental set-up will be used for a more detailed comparisons on a real battery.

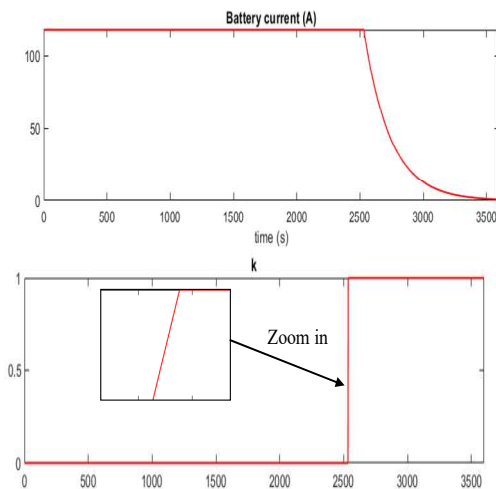


Fig. 12. Second algorithm: battery current and distribution input

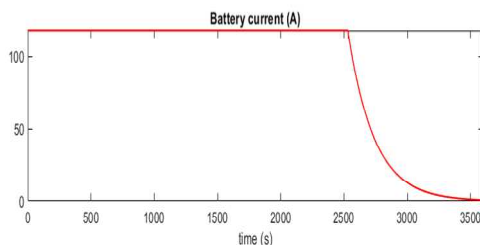


Fig. 13. Third algorithm: battery current

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REFERENCES

- [1] H. Liu et al., « An analytical model for the CC-CV charge of Li-ion batteries with application to degradation analysis », *Journal of Energy Storage*, vol. 29, p. 101342, juin 2020, doi: 10.1016/j.est.2020.101342.
- [2] E. Banguero, A. Correcher, Á. Pérez-Navarro, F. Morant, and A. Aristizabal, 'A Review on Battery Charging and Discharging Control Strategies: Application to Renewable Energy Systems', *Energies*, vol. 11, no. 4, Art. no. 4, Apr. 2018, doi: [10.3390/en11041021](https://doi.org/10.3390/en11041021).
- [3] B. Bose, A. Garg, B. K. Panigrahi, et J. Kim, « Study on Li-ion battery fast charging strategies: Review, challenges and proposed charging framework », *Journal of Energy Storage*, vol. 55, p. 105507, nov. 2022, doi: 10.1016/j.est.2022.105507.
- [4] W. Liu, T. Placke, and K. T. Chau, 'Overview of batteries and battery management for electric vehicles', *Energy Reports*, vol. 8, pp. 4058–4084, Nov. 2022, doi: [10.1016/j.egy.2022.03.016](https://doi.org/10.1016/j.egy.2022.03.016).
- [5] N. Ghaeminezhad et M. Monfared, « Charging control strategies for lithium-ion battery packs: Review and recent developments », *IET Power Electronics*, vol. 15, n° 5, p. 349-367, 2022, doi: 10.1049/pel2.12219.
- [6] R. German, P. Delarue and A. Bouscayrol, "Battery pack self-heating during the charging process," 2018 IEEE International Conference on Industrial Technology (ICIT), Lyon, France, 2018, pp. 2049-2054, doi: 10.1109/ICIT.2018.8352504.
- [7] A. Bouscayrol, B. Lemaire-Semail, "Energetic Macroscopic Representation and Inversion-Based Control ", *Encyclopedia of electrical and electronic power engineering*, Vol. 3, pp 365-375, Elsevier, DOI : 10.1016/B978-0-12-821204-2.00117-3, ISBN : 978-012-823211-8, 2023.
- [8] M. Brenna, F. Foiadelli, C. Leone, and M. Longo, 'Electric Vehicles Charging Technology Review and Optimal Size Estimation', *J. Electr. Eng. Technol.*, vol. 15, no. 6, pp. 2539–2552, Nov. 2020, doi: [10.1007/s42835-020-00547-x](https://doi.org/10.1007/s42835-020-00547-x).
- [9] G.-J. Chen and W.-H. Chung, 'Evaluation of Charging Methods for Lithium-Ion Batteries', *Electronics*, vol. 12, no. 19, Art. no. 19, Jan. 2023, doi: [10.3390/electronics12194095](https://doi.org/10.3390/electronics12194095).
- [10] P. Barrade, A. Bouscayrol, P. Delarue, "An energetic based method leading to merged control loops for the stability of input filters", *IEEE-VPPC'10*, Lille (France), September 2010
- [11] R. R. Kumar, C. Bharatiraja, K. Udhayakumar, S. Devakirubakaran, K. S. Sekar and L. Mihet-Popa, "Advances in Batteries, Battery Modeling, Battery Management System, Battery Thermal Management, SOC, SOH, and Charge/Discharge Characteristics in EV Applications," in *IEEE Access*, vol. 11, pp. 105761-105809, 2023, doi: 10.1109/ACCESS.2023.3318121.

Appendix A: EMR pictograms

	Source element (energy source)		Accumulation element (energy storage)		Indirect inversion (closed-loop control)
	Mono-physical conversion element		Mono-physical coupling element (energy distribution)		Direct inversion (open-loop control)
	Multi-physical conversion element		Multi-physical coupling element (energy distribution)		Coupling inversion (energy criteria)